

Tian Qiu

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EDUCATION

Rensselaer Polytechnic Institute

Ph.D. in Electrical Engineering

Troy, NY, USA

Sep. 2023 – Present

- **GPA:** 4.0/4.0

Huazhong University of Science and Technology

M.Eng. in Electrical Engineering

Wuhan, China

Sep. 2020 – Jun. 2023

- **GPA:** 88.32/100

Huazhong University of Science and Technology

B.Eng. in Electrical Engineering and Automation

Wuhan, China

Sep. 2016 – Jun. 2020

- **GPA:** 3.93/4.0 (Rank: 13/186)

PROFESSIONAL SKILLS

- **Semiconductor Characterization:** GaN HEMT Dynamic $R_{DS(on)}$ Characterization, Double-Pulse/Multi-Pulse Testing (DPT), Cryogenic & High-Temperature Testing.
- **Automated Test Equipment (ATE):** Development of 24/7 Automated Test Systems, Instrument Control (Oscilloscope, DC Source) via MATLAB, High-Throughput Data Acquisition.
- **Hardware & PCB Design:** Cryogenically Reliable Gate Drivers, Voltage Clipper Probe, Current Shunt Sensor, Power Stage Design, Layout Optimization (Altium Designer).
- **Embedded Control:** DSP Firmware Development (TI CCS), Flexible PWM Generation, Digital Control Interface Design.
- **Modeling & Mechanical:** Device Modeling (Neural Network/Analytical), Cryogenic Vacuum Chamber Design, Thermal Management, 3D Modeling (SolidWorks).

RESEARCH EXPERIENCE

Rensselaer Polytechnic Institute

Research Assistant — Advisor: Prof. Z. Zhang

Sep. 2023 – Present

Project: Automated Cryogenic Characterization Platform for GaN Power Semiconductors

- Designed and deployed a **fully automated, high-throughput characterization system** for comprehensive dynamic $R_{DS(on)}$ assessment of GaN HEMTs, enabling **24/7 autonomous operation** with a data yield of **3,000+ sets per day**.
- **Hardware Innovation:** Developed and optimized **measurement tools** including a high-bandwidth voltage clipper probe and current sensor; designed a **cryogenic gate driver**, a power stage motherboard with fiber-optic communication, and optimized cryogenic system for extreme temperature reliability (50 K – 400 K).
- **Control & Software:** Programmed **DSP** firmware to generate flexibly controlled double/multi-pulse signals; developed a **MATLAB-based automation framework** to coordinate DC sources and oscilloscopes, enabling auto-sweeping of voltage, current, switching modes, and temperature conditions.
- **Protection:** Proposed an **ultrafast** short-circuit protection scheme for GaN HEMTs based on drain-source voltage pattern recognition, featuring a detection delay **< 50 ns**.
- **Modeling & Analysis:** Synthesized large-scale experimental datasets to establish a **data-driven device model** using neural networks.

Huazhong University of Science and Technology

Research Assistant — Advisor: Prof. H. Li

Oct. 2019 – Jun. 2023

Project: Performance degradation study of metallized film capacitors under harsh environments

- Established an online monitoring platform to study degradation behaviors under atmospheric and electrochemical aging.
- Developed analytical models for nano-electrode recession, ESR elevation, capacitance degradation and lifetime prediction at high-temperature & high-humidity conditions.

SELECTED PUBLICATIONS

Journal Articles (Selected)

1. **T. Qiu**, Z. Zhang, P. Khadka, et al., “Modeling and Design Method of Interconnection for Cryogenic Power Electronics Testing Down to 50 K,” *submitted to IEEE Journal of Emerging and Selected Topics in Power Electronics*, 2026.

2. H. Li, **T. Qiu**, et al., “ESR Modeling for Atmospheric Corrosion Behavior of Metallized Film Capacitors,” *IEEE Transactions on Device and Materials Reliability*, 2023.
3. H. Li, **T. Qiu**, et al., “Capacitance loss evolution of Zn–Al metallized film capacitors under atmospheric corrosion,” *IEEE Transactions on Dielectrics and Electrical Insulation*, 2022.
4. **T. Qiu**, H. Li, et al., “A proposed RT model for atmospheric corrosion of metallized film in AC capacitors,” *IEEE Transactions on Dielectrics and Electrical Insulation*, 2022.

Conference Proceedings (Selected)

5. **T. Qiu**, Z. Zhang, et al., “Characterization and Comparison of Isolated Gate Drivers Considering Diverse Techniques for Cryogenic Applications,” in *IEEE Applied Power Electronics Conference (APEC)*, 2026.
6. **T. Qiu**, Z. Zhang, I. Zikpi and A. Das, “Ultrafast Noise-Immune Short-Circuit Detection in Gallium Nitride Power Semiconductors via Drain-Source Voltage Pattern Recognition,” in *IEEE Energy Conversion Congress and Exposition (ECCE)*, 2025.
7. **T. Qiu**, Z. Zhang, et al., “Automated Characterization Platform for Comprehensive Dynamic Rdson Assessment of GaN HEMTs From 50 K to 400 K,” in *IEEE APEC*, 2025.
8. **T. Qiu**, Z. Zhang, et al., “Conduction Lead Design and Optimization for Cryogenic Characterization of GaN HEMTs,” in *IEEE ECCE*, 2024.

CONFERENCE DEMONSTRATIONS

- **IEEE APEC Student Project Demonstration** (Accepted) Mar. 2026
Project in Preparation
- **IEEE ECCE Student Project Demonstration** Oct. 2025
Project: Reliability Characterization Platform with In-Circuit Characterization and Online Monitoring of GaN HEMTs
- **IEEE ECCE Student Project Demonstration** Oct. 2024
Project: Characterization Automation for GaN HEMTs from 50 K to 400 K for Space Power

HONORS & AWARDS

- IEEE PELS Best ECCE Oral Presentation Paper 2024
- Master Student Scholarship 2021, 2022
- Contemporary Undergraduate Mathematical Contest in Modeling: First Prize 2018
- National University Students Electrical Math Modeling Competition: Third Prize 2018